

Referential readings of reflexives

Abstract

I show that reflexives in DP-sized Right Node Raising pivots can have strict readings, just like pronouns in the same environment. Certain properties of these examples preclude an elliptical analysis of Right Node Raising. A multidominant analysis of these examples follows if we assume strict readings come from a referential reading of these reflexives and pronouns. This paper thus supports a treatment of reflexives that distinguish referential from bound readings. Reflexives are like pronouns, except they have to satisfy Condition A.

Keywords: reflexives, Right Node Raising, multidominance, Condition A

Pronouns and reflexives are closely related forms of anaphora.¹ Many theories of the syntax-semantics interface take them to differ in one important aspect: while pronouns are ambiguous between referential readings and bound variable readings, reflexives aren't: they only have bound variable readings (Partee and Bach 1984; Williams 1977). This paper calls this treatment into question: do reflexives have referential readings after all?

In this paper, I show that Right Node Raising constructions with DP-sized pivots feature strict readings of reflexives:

- (1) John LOVES, but Bill HATES, pictures of himself.

Right Node Raising is often thought to have at least two derivational sources: one elliptical (Abels 2004; Bošković 2004/1996; Giannakidou and Merchant 1998; Ha 2006, 2008; Hartmann 2000; Wilder 1997), and one multidominant (Bachrach and Katzir 2007a,b; Gračanin-Yuksek 2007; Grosz 2009; McCawley 1982; Vos and Vicente 2005; Wilder 1999). I argue that an elliptical analysis of Right Node Raising with DP-sized pivots is implausible. Adopting a multidominant analysis of Right Node Raising, I show that the strict reading of reflexives is accounted for once we assume that reflexives have referential readings, which in turn offers support for adopting this assumption more generally. Thus, reflexives have both referential and bound readings, just like pronouns.

It's well-known that reflexives have strict readings under ellipsis (Ahn and Sportiche 2014; Buring 2005; Dahl 1973; Fiengo and May 1994; Hestvik 1995; I. Sag 1976). This fact is often used in arguments that make the very same case that reflexives have referential readings (McKillen 2016). However, these arguments are complicated by the presence of an alternative ellipsis account that derives strict readings from bound variable interpretations of reflexives. Thus, it's important to rule out an elliptical account of the key Right

Node Raising examples in this paper, so that the same complications don't plague our argument.

This paper is structured as follows. In Section 1, I review the argument for referential reflexives that follows from strict readings of reflexives under ellipsis. I also discuss Fox's (2000) alternative ellipsis account that defuses this argument. In Section 2, I argue that Right Node Raising constructions with DP-sized pivots aren't derived by ellipsis, while multidominance remains an analytical option. In Section 3, I introduce a set of Right Node Raising facts that show what range of readings pronouns and reflexives in DP-sized pivots can have, including (1). It turns out that pronouns and reflexives feature almost the same range of strict and sloppy readings. The only caveat is that reflexives don't have a reading where they don't have an antecedent in the sentence, a Condition A effect that is utterly expected. I show that both the pronoun and reflexive facts receive a multidominant analysis once we assume pronouns and reflexives both have distinct referential and bound readings. I conclude in Section 4.

1 Background: strict reading of reflexives under ellipsis

One argument for assuming that reflexives have referential readings comes from the fact that reflexives can have strict readings under ellipsis (Ahn and Sportiche 2014; Büring 2005; Dahl 1973; Fiengo and May 1994; Hestvik 1995; I. Sag 1976). For example, (2) is ambiguous between a strict vs. sloppy reading of the reflexive *himself*.

- (2) John defended himself better than BILL did.
- a. *Strict*
John defended John better than Bill defended John.
 - b. *Sloppy*
John defended John better than Bill defended Bill.

All theories of VP Ellipsis require the elided VP to be recoverable from an antecedent,² even though they differ with respect to the exact nature of this recoverability condition (Fox 2000; Heim 1997; Merchant 2001; I. Sag 1976; Williams 1977, many others). Under most theories of ellipsis, an account of the sloppy readings of (2) straightforwardly follows from the assumption that the reflexive in the antecedent VP denotes a bound variable. For example, (2b), the sloppy reading of (2), might have the structure (3).

- (3) *Structure for (2b)*
- $$[_A \text{ John } \lambda 1 t_1 \text{ defended himself}_1] [_{\text{better than}} \text{ } [_E \text{ F[BILL] } \lambda 2 t_2 \text{ did defend himself}_2]].$$

Let's see how ellipsis is licensed in (3). For concreteness, I will assume that VP Ellipsis is licensed by a semantic identity condition (Heim 1997; Rooth 1992) constrained by a parallelism condition (Fiengo and May 1994; Fox 1995, 2000). The semantic identity condition (4a) says there must be a pair of constituents, which reflexively dominate respectively

the elided VP and its antecedent, and have equal denotations except for any focus-marked parts. Parallelism, a constraint on variable binding, is given in (4b). Let's refer to linguistic expressions that denote variables, no matter free or bound, as *syntactic variables*; this includes traces, pronouns, reflexives, etc. Parallelism requires syntactic variables in the elided VP and its antecedent to enter into parallel binding dependencies.

(4) *VP Ellipsis licensing condition*

For an elided VP, VP_E , there must be an antecedent VP, VP_A , such that:

a. *Semantic identity*

VP_E and VP_A are reflexively dominated by constituents E and A respectively, such that $\llbracket A \rrbracket^g$ is a member of $F(E)$, the set of focus alternatives of E .

b. *Parallelism*

Each bound syntactic variable in VP_E is parallel with something in VP_A , and each bound syntactic variable in VP_A is parallel with something in VP_E .

If X and Y are phrases, then X is *parallel* with Y iff

- i. X and Y are bound from parallel positions.

In (3), the clauses containing the antecedent VP and the elided VP respectively satisfy semantic identity (4a), as detailed in (5). The elided reflexive and its counterpart in the antecedent VP are both bound syntactic variables. Since they are bound from subject positions, Parallelism (4b) is also satisfied. Thus, the sloppy reading is licensed.

(5) *Semantic identity satisfaction in (3)*

$$\begin{aligned}\llbracket A \rrbracket^g &= \llbracket \text{John } \lambda 1 t_1 \text{ defended himself}_1 \rrbracket^g \\ &= \text{defended}(\text{john})(\text{john}) \\ F(E) &= F(\text{F[BILL]} \lambda 2 t_2 \text{ defended himself}_2) \\ &= \{\text{defended}(x)(x) \mid x \in D_e\}\end{aligned}$$

Thus, $\llbracket A \rrbracket^g \in F(E)$.

However, the strict reading of (2) can't be licensed if the reflexive in the antecedent VP is a subject-bound variable. This is because Parallelism demands that the elided VP also contain a subject-bound variable. That gives rise to the sloppy reading (2b), not the strict one (2a). Thus, to account for the strict reading, we need reflexives that aren't bound variables (McKillen 2016).

Such an account is possible if we assume reflexives have referential readings. Effectively, this means we treat referential reflexives as referential pronouns that additionally satisfy Condition A. We need to formulate Condition A so that it constrains both coreference and binding. This is done in (8), with familiar locality conditions in place (Chomsky 1980, 1981).

(6) *Smallestness*

Let S be a set of phrases, and $X \in S$. Then, X is *smallest* in S iff there is no $Y \in S$ such that X dominates Y but Y doesn't dominate X .

(7) *Binding domain*

Let R be a reflexive. A smallest phrase that dominates R and also dominates an external argument or a possessor that c-commands R is a *binding domain* for R .

(8) *Condition A (McKillen 2016, p. 85, ex. 3)*

A reflexive R must be bound by or coreferential with a constituent Y , which is called the *antecedent* of R , such that

- a. there is no binding domain of R that doesn't dominate Y ,
- b. Y c-commands R , and
- c. Y is in an A-position.

Let's follow Heim and Kratzer's (1998) treatment of referential pronouns, which take them to be variables that aren't syntactically lambda-bound. A structure for the strict reading (2a) can then be (9): the elided object and its antecedent are both indexed 3, bound by neither conjunct's subject. I assume vehicle change (Fiengo and May 1994) has taken place in the elided VP, allowing the elided object to take a pronominal form. Otherwise, the elided object would have to be a reflexive that lacks a local antecedent, hence violating Condition A.

(9) *Structure for (2a)*

John $\lambda 1 t_1$ defended himself₃ [better than [_F[BILL] $\lambda 2 t_2$ did defend him₃]].

Ellipsis is correctly predicted to be licensed. Parallelism (4b) is satisfied vacuously because neither object in the two VPs is a bound syntactic variable. Semantic identity (4a) is again satisfied by the same two clauses labelled A and E , as shown in (10). The key assumption is that $g(3) = \text{john}$. We can ignore the cases where $g(3) \neq \text{john}$, because then the reflexive *himself*₃ is neither bound by nor corefers with anything, violating Condition A in any sensible formulation, thus crashing the derivation.

(10) *Semantic identity satisfaction in (9)*

Assume $g(3) = \text{john}$.

$$\begin{aligned} \llbracket A \rrbracket^g &= \llbracket \text{John } \lambda 1 t_1 \text{ defended himself}_3 \rrbracket^g \\ &= \text{defended}(\text{john})(\text{john}) \\ F(E) &= F(\llbracket \text{BILL} \rrbracket \lambda 2 t_2 \text{ defended him}_3) \\ &= \{\text{defended}(\text{john})(x) \mid x \in D_e\} \end{aligned}$$

Thus, $\llbracket A \rrbracket^g \in F(E)$.

Reflexives with referential readings are necessary only to the extent that the account we just described is uniquely successful in accounting for the strict reading of reflexives. However, Fox (2000) has an alternative account that generates the strict reading from an antecedent VP containing a bound variable reflexive. His condition, which we may call Alternative Parallelism (11), relaxes Parallelism by introducing a new clause (11b).

(11) *Alternative Parallelism (adapted from Fox 2000, p. 117)*

Each bound syntactic variable in VP_E is parallel with something in VP_A , and each bound syntactic variable in VP_A is parallel with something in VP_E .

If X and Y are phrases, then X is *parallel* with Y iff

- a. X and Y are bound from parallel positions; or
- b. X and Y end up having the same referential values.

Then, the strict reading (2a) may have the structure in (12). The reflexive in the antecedent, *himself₁*, is a variable bound by *John*. Its counterpart in the elided VP could be a pronoun – an effect of vehicle change – which is referential.

(12) *Another structure for (2a)*

John $\lambda 1 t_1$ defended himself₁ [better than [_F[BILL] $\lambda 2 t_2$ did defend him₃]].

This structure passes the new ellipsis licensing condition which uses Alternative Parallelism (11). Assume $g(3) = \text{john}$. Even though the reflexive in the antecedent VP and the pronoun in the elided VP don't enter into parallel binding dependencies, they end up resolving to the same person, John. On this basis, they count as parallel by (11b), thus satisfying Alternative Parallelism. Semantic identity (4a) is checked as well.

Fox's account successfully generates the strict reading without ever assuming reflexives can have a referential reading. As such, our need to posit referential reflexives is defused to the extent that Fox's account is empirically more successful than the previous account. The reality is that there is no clear winner between the two accounts (see Fleisher 2023 for a recent summary of the debate). Thus, it is fair to say that strict readings of reflexives under ellipsis don't offer clear support for referential reflexives.

2 Right Node Raising with DP-sized pivots aren't derived by ellipsis

In the next section, I will focus on Right Node Raising examples with DP-sized pivots, and show that these examples feature reflexives with strict readings. These facts support positing referential reflexives. Right Node Raising has received at least two kinds of analysis: elliptical (Abels 2004; Bošković 2004/1996; Giannakidou and Merchant 1998; Ha 2006, 2008; Hartmann 2000; Wilder 1997), and multidominant (Bachrach and Katzir 2007a,b; Gračanin-Yuksek 2007; Grosz 2009; McCawley 1982; Vos and Vicente 2005; Wilder 1999). In an elliptical analysis of the Right Node Raising example (13), the pivot

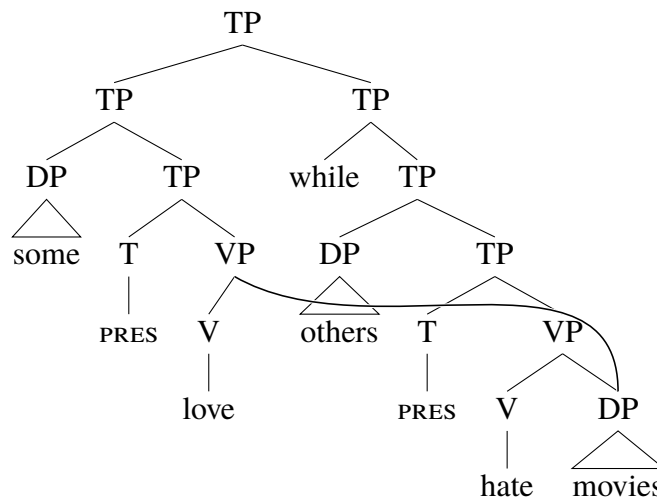
is only contained in the second conjunct, and it antecedes the ellipsis of its counterpart in the first conjunct (14a). In a multidominant analysis (14b), the pivot is a single syntactic object multidominated within the two conjuncts. While some take Right Node Raising to be exclusively generated by one of these analytical options, others adopt a hybrid approach whereby both derivational sources are available in the grammar (Barros and Vicente 2011).

(13) Some love, while others hate, movies.

(14) a. *Elliptical derivation for (13)*

Some love ~~movies~~, while others hate movies.

b. *Multidominant derivation for (13)*



I will adopt a two-column representation for multidominant structures, which collapses (14b) to (15). The material on the second column is the pivot, and the material on the two rows of the first column are the conjuncts in which the pivot is multidominated.

(15) *Same as (14b)*

Some [love,	movies].
while others [hate,	

In the previous section, we learned that strict readings of reflexives under ellipsis can be accounted for without making them referential. Such an account will haunt our subsequent discussion, given that ellipsis is an *a priori* possible derivational source for Right Node Raising. In this section, I offer arguments for eliminating the elliptical analysis, at least for Right Node Raising constructions with DP-sized pivots, and suggest that we adopt a multidominant analysis for them.

My first argument is specific to Right Node Raising with DP-sized pivots. English doesn't allow DPs to be elided in general (16), so ellipsis can't create DP-sized pivots.

(16) *English doesn't allow DP Ellipsis*

Sally only tasted my pumpkin pie. *She didn't eat ~~my pumpkin pie~~.

My second argument is more general: the multidominant account is uniquely successful in deriving the Right Edge Restriction (Wilder 1997), a generalization that imposes a linear order constraint on Right Node Raising. Put simply, the Right Edge Restriction states that the pivot's counterpart in the first conjunct must be final in that conjunct. Its effects are illustrated in (17), where Right Node Raising occurs at VP-level conjunctions. In (17a), the pivot is *the book*. Its counterpart in the first VP *fetch* is final in this VP, since it is the direct object. Thus, (17a) is ruled in. In (17b), the pivot is *that girl*. Its counterpart in the first VP *give the book* is not final in this VP, since it is the indirect object in a double object construction. Thus, (17b) is ruled out.

(17) *Right Edge Restriction examples (adapted from Wilder 1999)*

- a. John should [_{VP} [_{VP} fetch] and [_{VP} give the book to Mary]].
(Intended: John should fetch the book and give the book to Mary.)
- b. * John should [_{VP} [_{VP} give the book] and [_{VP} praise that girl]].
(Intended: John should give that girl the book and praise that girl.)

On the other hand, two properties of Right Node Raising examples have been taken to be exclusively predicted by an elliptical analysis: sloppy readings of pronouns and tolerance of morphological mismatch. I will show that sloppy identity isn't a reliable diagnostic for ellipsis, as a multidominated pronoun can have a sloppy reading, too. As for morphological mismatch, I will show that it doesn't occur with DP-sized pivots, hence suggesting that these Right Node Raising constructions are derived from multidominance, not ellipsis.

First, consider sloppy readings. Right Node Raising is known to feature sloppy readings of pronouns (Dahl 1981; Höhle 1990; Jacobson 1999; Stechow 1990), even in DP-sized pivots like (18).

(18) *Sloppy identity in Right Node Raising with DP-sized pivots (Ha 2008, p. 87)*
Mary LOVED, but Sue HATED, her twin sister.

As we saw in Section 1, ellipsis also admits sloppy readings. Sloppy identity is sometimes taken to be a diagnostic for ellipsis, but it is actually a diagnostic for binding (Merchant 2013). Multidominance can also account for sloppy identity if we take the sloppy pronoun to be a bound variable that is bound in both conjuncts (19) (Holladay 2023). Its interpretation is given in (20).

(19) *Multidominant account of (18)*

Mary λ1 t ₁ [LOVED,	her ₁ twin sister].
but Sue λ1 t ₁ [HATED,	

$$\begin{aligned}
(20) \quad \llbracket (19) \rrbracket &= \llbracket \text{but} \rrbracket (\llbracket \text{Mary } \lambda 1 t_1 [\text{LOVED her}_1 \text{ twin sister}] \rrbracket) \\
&\quad (\llbracket \text{Sue } \lambda 1 t_1 [\text{HATED her}_1 \text{ twin sister}] \rrbracket) \\
&= \text{but}((\lambda x_e. \text{loved}(\text{twin-sister}(x))(x))(\text{mary})) \\
&\quad ((\lambda y_e. \text{hated}(\text{twin-sister}(y))(y))(\text{sue})) \\
&= \text{but}(\text{loved}(\text{twin-sister}(\text{mary}))(\text{mary})) \\
&\quad (\text{hated}(\text{twin-sister}(\text{sue}))(\text{sue}))
\end{aligned}$$

Next, consider morphological mismatch, which happens when an element in the pivot is supposed to establish an agreement dependency with some element in each conjunct. When the two dependencies have conflicting demands for the morphological form of the element in the pivot, the demand from the second conjunct must be honored; a morphological mismatch with the first conjunct is apparently tolerated. In (21), the auxiliary *has* in the first conjunct requires the form of the verb heading the VP pivot to be *slept*, but the modal *will* in the second conjunct requires it to be *sleep*. The conflicting requirements are apparently resolved in favor of the second conjunct.

- (21) *Morphological mismatch in Right Node Raising (Bošković 2004/1996)*
 John has, and Bill definitely will, { sleep / *slept } on Sally's couch.

This is expected in an elliptical derivation (22): the first conjunct has its own elided counterpart of the pivot where its morphological demand is met. On the other hand, such morphological mismatch shouldn't be tolerated in a multidominant derivation, because a shared element cannot enter into conflicting morphological dependencies.

- (22) *Elliptical account of (21)*
 John has ~~slept on Sally's couch~~, and Bill definitely will sleep on Sally's couch.

The pair (23) shows that such morphological mismatch is much less tolerable in Right Node Raising with DP-sized pivots (23b) than it is in VP Ellipsis (23a). Sloppy identity is possible in (23a) but not in (23b); (23b) only has a strict reading. This contrast is unexpected if these cases of Right Node Raising admit an elliptical derivation.

- (23) a. I love my family, and you do ~~love your family~~, too.
 b. *? You LOVE, but I HATE, my family.
 (Intended: you love your family, but I hate my family.)

To summarize, I have argued against an elliptical analysis for Right Node Raising with DP-sized pivots, which we exclusively focus on for the rest of this paper.³

Can we assume a multidominant analysis? Yes, but a note is in order. Yet another analytical option for Right Node Raising is to generate the pivot with rightward Across-the-Board (ATB) movement (Clapp 2008; Postal 1998; Ross 1967; Sabbagh 2007, 2008). I won't explore this option, because an ATB-movement analysis of Right Node Raising is quite compatible with a multidominant analysis, especially in light of multidominant implementations of ATB-movement (Citko 2005). The distinction between the predictions

made by the two analyses of Right Node Raising is accordingly difficult to locate. Therefore, I will not pursue this issue further in this paper.

3 What kind of readings do pronouns and reflexives have in a DP-sized Right Node Raising pivot?

In this section, I show that reflexives in DP-sized Right Node Raising pivots have strict readings, just like pronouns. This means reflexives and pronouns have almost exactly the same range of readings in Right Node Raising with DP-sized pivots, except reflexives show a Condition A effect. If we assume these Right Node Raising constructions have a multidominant source, it is difficult to account for this fact unless we distinguish referential and bound readings of reflexives, just like we do for pronouns. To the extent that the multidominant account is necessary, we have an argument to adopt the general assumption that reflexives can have referential in addition to bound readings.

This in fact mirrors the argument for the same assumption from strict readings of reflexives under ellipsis. As we reviewed in Section 1, that argument is complicated by an alternative ellipsis account that avoids positing referential reflexives. However, we have eliminated in Section 2 an elliptical analysis of the Right Node Raising examples under consideration, which have DP-sized pivots. Thus, this paper makes a stronger argument for the presence of referential reflexives.

Let us first consider the range of readings a pronoun in a DP-sized Right Node Raising pivot may have. Consider (24). The pronoun *his* can establish strict identity in three ways: with the first conjunct's subject (25a), with the second conjunct's subject (25b), or with a contextually salient entity that is neither conjunct's subject (25c). It can also have a sloppy reading (25d).

(24) John LOVES, but Bill HATES, his father.

(25) a. *Strict with first*

John loves John's father, but Bill hates John's father.

b. *Strict with second*

John loves Bill's father, but Bill hates Bill's father.

c. *Strict with neither*

John loves X's father, but Bill hates X's father.

(where X is a contextually salient man.)

d. *Sloppy*

John loves John's father, but Bill hates Bill's father.

A popular idea is that strict readings arise from referential pronouns, and sloppy readings from bound ones (I. Sag 1976; Williams 1977). Then, the range of possible readings of the pronoun *his* in (24) seems to be described by the generalization in (26).

- (26) A pronoun in a DP-sized pivot of a (conjunctive) Right Node Raising structure must be referential in both conjuncts, or bound in both conjuncts.

By (26), we correctly rule out (27a) and (27b), which are *mixed* readings of the pronoun in (24). In these readings, the pronoun *his* is subject-bound in one conjunct but refers to a contextually salient entity in the other conjunct. .

- (27) a. *Mixed: bound in first, referential in second*
 John loves John's father, but Bill hates X's father.
 (where X is a contextually salient man.)
 b. *Mixed: referential in first, bound in second*
 John loves X's father, but Bill hates Bill's father.
 (where X is a contextually salient man.)

A multidominant analysis can derive the generalization (26) only by making referential and bound pronouns denote distinct semantic objects (Reinhart 1983a,b). For concreteness, assume referential and bound pronouns both denote variables, identified by their indices (Heim and Kratzer 1998). Then, we need to prevent a referential pronoun and a bound one in the same sentence from using the same index. I will describe one specific implementation of this idea, where I designate different kinds of indices for referential and bound pronouns. Bound pronouns carry *bound indices*, which are drawn as usual from the positive integers $\mathbb{N} = \{1, 2, \dots\}$. Referential pronouns carry *referential indices*, which I will notate with hats; they are thus drawn from the set of hatted positive integers $\hat{\mathbb{N}} = \{\hat{1}, \hat{2}, \dots\}$. An assignment function can contain mappings for both kinds of indices. Thus, both kinds of pronouns can be interpreted with Heim and Kratzer's (1998) Traces and Pronouns rule (28).

- (28) *Traces and Pronouns (adapted from Heim and Kratzer 1998)*
 Let g be an assignment function, $k \in \mathbb{N} \cup \hat{\mathbb{N}}$ a (bound or referential) index and x a trace or a pronoun. Then $\llbracket x_k \rrbracket^g = g(k)$.

We can't allow referential and bound pronouns to behave alike. We need to make Heim and Kratzer's Predicate Abstraction (29) rule apply only to bound indices. This prevents referential pronouns from being bound. To prevent bound pronouns from being referential, we need to restrict assignment functions used to interpret sentences so that they don't contain any mappings for bound indices (30).

- (29) *Predicate Abstraction (adapted from Heim and Kratzer 1998)*
 Let g be an assignment function, $k \in \mathbb{N}$ a bound index and α a tree with a defined denotation $\llbracket \alpha \rrbracket^g$. Then

$$\llbracket \lambda_k \alpha \rrbracket^g = \lambda x_e. \llbracket \alpha \rrbracket^{g[x/k]}$$

- (30) *Restriction on assignment functions*
 A sentence can be interpreted with respect to an assignment function g only if $g(k)$ is undefined for any bound index $k \in \mathbb{N}$.

Then, the four grammatical readings of (24) receive the following structures, with appropriate restrictions for the assignment function used to interpret them in place.

(31) a. *Multidominant derivation of (25a)*

John $\lambda 1 t_1$ [loves,	his ₃ father].
but Bill $\lambda 2 t_2$ [hates,	

(Interpret this with a g such that $g(\hat{3}) = \text{john}$).

b. *Multidominant derivation of (25b)*

John $\lambda 1 t_1$ [loves,	his ₃ father].
but Bill $\lambda 2 t_2$ [hates,	

(Interpret this with a g such that $g(\hat{3}) = \text{bill}$).

c. *Multidominant derivation of (25c)*

John $\lambda 1 t_1$ [loves,	his ₃ father].
but Bill $\lambda 2 t_2$ [hates,	

(Interpret this with a g such that $g(\hat{3}) = \text{a contextually salient man.}$)

d. *Multidominant derivation of (25d)*

John $\lambda 1 t_1$ [loves,	his ₁ father].
but Bill $\lambda 1 t_1$ [hates,	

This system also rules out the ungrammatical readings in (27). Take example (27a), where the pronoun *his* is bound in the first conjunct but referential in the second conjunct. The only ways to generate this reading are to either lambda-bind a referential pronoun (32a) or to map a bound index to a contextually salient entity in the assignment function (32b). But our system allows neither structure; (32a) can't be interpreted via Predicate Abstraction (29), while (32b) violates the restriction (30). Had we used the same set of indices for referential and bound pronouns, we would incorrectly generate this unattested reading.

(32) *Impossible multidominant derivations of (27a)*

a.

John $\lambda \hat{1} t_{\hat{1}}$ [loves,	his ₁ father].
Bill $\lambda 2 t_2$ [hates,	

(Interpret this with a g such that $g(\hat{1}) = \text{a contextually salient man.}$)

b.

John $\lambda 1 t_1$ [loves,	his ₁ father].
Bill $\lambda 2 t_2$ [hates,	

(Interpret this with a g such that $g(1) = \text{a contextually salient man.}$)

My system is just one way to derive (26). In particular, Heim and Kratzer's (1998) condition (33) and Heim's (1997) *No Meaningless Coindexing* (34) work equally well.⁴

- (33) No LF representation (for a sentence or multisentential text) must contain both bound occurrences and free occurrences of the same index. (Heim and Kratzer 1998, p. 254)
- (34) *No Meaningless Coindexing* (Heim 1997)
 If an LF contains an occurrence of a variable v that is bound by a node α , then all occurrences of v in this LF must be bound by the same node α .

These are syntactic constraints on indexing. To derive (26), they depend on a notion of *occurrences* that distinguishes multiple occurrences of a multidominated syntactic object (Chomsky 2000; Collins and Stabler 2016; Kracht 1999; Wilder 2008). My implementation doesn't make use of syntactic constraints and thus doesn't have this dependency.

Having understood the range of readings pronouns have, we now turn to the true subject of this paper: reflexives in DP-sized Right Node Raising pivots (1). What readings do they have? Suppose reflexives only have bound readings. We would then predict only a sloppy reading to be possible for the reflexive *himself* in (1). Contrary to our prediction, strict readings are available (35a), (35b), in addition to a sloppy reading (35d).

- (1) John LOVES, but Bill HATES, pictures of himself.
- (35) a. *Strict with first*
 John loves pictures of John, but Bill hates pictures of John.
- b. *Strict with second*
 John loves pictures of Bill, but Bill hates pictures of Bill.
- c. *Strict with neither*
 *John loves pictures of X, but Bill hates pictures of X.
 (X is a contextually salient man.)
- d. *Sloppy*
 John loves pictures of John, but Bill hates pictures of Bill.

We are thus led to conclude that reflexives do have referential readings, at least if a multidominant account of these examples is assumed.

The three grammatical readings of (1) receive the following structures, with appropriate restrictions for assignment functions in place.

- (36) a. *Multidominant derivation of (35a)*
- | | |
|-----------------------------------|-------------------------------------|
| John $\lambda 1 t_1$ [loves, | pictures of himself ₃]. |
| but Bill $\lambda 2 t_2$ [hates, | |
- (Interpret this with a g such that $g(\hat{3}) = \text{john}$).
- b. *Multidominant derivation of (35b)*
- | | |
|-----------------------------------|-------------------------------------|
| John $\lambda 1 t_1$ [loves, | pictures of himself ₃]. |
| but Bill $\lambda 2 t_2$ [hates, | |
- (Interpret this with a g such that $g(\hat{3}) = \text{bill}$).

c. *Multidominant derivation of (35d)*

John $\lambda 1 t_1$ [loves,	pictures of himself ₁].
but Bill $\lambda 1 t_1$ [hates,	

The only reading (1) doesn't have is (35c), where the reflexive *himself* refers to a contextually salient entity that is neither conjunct's subject. This reading would be generated by the structure (37).

(37) *Multidominant derivation of (35c)*

John $\lambda 1 t_1$ [loves,	pictures of himself ₃].
but Bill $\lambda 2 t_2$ [hates,	

(Interpret this with a g such that $g(\hat{3}) = \text{a contextually salient man.}$)

This structure is ruled out by Condition A, because *himself*₃ is not bound by or coreferential with anything in (37). In other words, it doesn't have an antecedent. But how is Condition A satisfied in the strict and sloppy readings? In the sloppy reading (36c), the reflexive has two antecedents, one in each conjunct. In the strict readings (36a), (36b), it only has one.

This state of affairs is in fact completely expected under Condition A as we formulated it in (8). The reflexives in (1) each have two binding domains, namely, the two conjuncts. The reflexive just needs to have an antecedent that satisfies Condition A in one conjunct; this is why strict readings are available. But why does a sloppy reflexive seem to necessarily have an antecedent in each conjunct? That's because a sloppy reflexive is a bound variable that isn't bound within the pivot. It thus needs a binder in each conjunct; otherwise, the reflexive ends up free in one conjunct, fatally violating (30).

The availability of strict readings of reflexives (35a), (35b) tells us that these reflexives only need to have a local antecedent in one conjunct. This has consequences for implementations of Condition A that take it to be a presupposition (McKillen 2016; Sauerland 2013) when it comes to figuring out what linguistic element carries the presupposition. Typically, the presupposition is thought to originate from the reflexive, as illustrated in the simple example (38), where the presupposition is indicated by the superscript PS. The Condition A presupposition projects up the clausal spine, enforcing the requirement for a local antecedent in this clause.

(38) John $\lambda 1 t_1$ loves pictures of himself₃^{PS}.

(Interpret this with a g such that $g(\hat{3}) = \text{john}$).

Then, the structure of (35a) would be (39).

(39) *Structure of (35a) under the presuppositional account of Condition A*

John $\lambda 1 t_1$ [loves,	pictures of himself ₃ ^{PS}].
but Bill $\lambda 2 t_2$ [hates,	

(Interpret this with a g such that $g(\hat{3}) = \text{john}$).

However, this predicts that the reflexive *himself* must find a local antecedent in both conjuncts, incorrectly ruling out the strict readings. This prediction arises because a presupposition in a multidominated Right Node Raising pivot projects in both conjuncts. For example, (40) presupposes that both John and Bill used to smoke:

- (40)
$$\begin{array}{c|c} \text{John } \lambda 1 \ t_1 \text{ considered,} & \\ \hline \text{and Bill already } \lambda 2 \text{ gave up,} & [\text{stopping smoking}]^{\text{PS}} \end{array}$$

Sauerland (2013) proposes that the presuppositional reflexive moves to a position right underneath its antecedent (41).

- (41)
$$\text{John } \lambda 1 \ t_1 \text{ himself}_3^{\text{PS}} \lambda 4 \text{ loves pictures of } t_4.$$

(Interpret this with a g such that $g(\hat{3}) = \text{john}$).

Our Right Node Raising examples of interest have object pivots. Sauerland's account allows the Condition A presupposition to be introduced above the object position, avoiding projection into both conjuncts. However, to generate the strict reading (35a) under his account, the reflexive needs to move into the first conjunct, where its antecedent is located. This movement creates a variable (t_4) in the pivot that is free in the second conjunct, rendering the structure uninterpretable.

- (42) *Structure of (35a) under Sauerland's (2013) account of Condition A*
$$\begin{array}{c|c} \text{John } \lambda 1 \ t_1 \text{ himself}_3^{\text{PS}} \lambda 4 [\text{loves,} & \\ \hline \text{but Bill } \lambda 2 \ t_2 [\text{hates,} & \text{pictures of } t_4 \end{array}$$

(Interpret this with a g such that $g(\hat{3}) = \text{john}$).

A presuppositional account of Condition A can generate the strict readings only by placing the Condition A presupposition outside the reflexive (citation redacted). For example, a reflexive might bear an uninterpretable feature that is checked by an element (e.g., finite T) that carries a Condition A presupposition. Then, the structure of (35a) would be (43).

- (43) *Structure of (35a) under an alternative presuppositional account of Condition A*
$$\begin{array}{c|c} \text{John } \lambda 1 \text{ PRES}^{\text{PS}} t_1 [\text{loves,} & \\ \hline \text{but Bill } \lambda 2 \ t_2 [\text{hates,} & \text{pictures of himself}_3 \end{array}$$

(Interpret this with a g such that $g(\hat{3}) = \text{john}$).

The finite present T head PRES in the first conjunct bears a Condition A presupposition, which is satisfied, and checks the uninterpretable feature off himself_3 . The result is grammatical. I leave the details of this analysis to other authors/future research.

We thus see that the reflexive example (1) patterns exactly like the pronoun example (24), except that the reflexive doesn't have a strict-with-neither reading. This teaches us that reflexives, like pronouns, have referential in addition to bound readings. In other words, reflexives are pronouns that happen to be constrained by Condition A.

I'd like to end with a puzzle. Consider (44), adapted from a reviewer's example.

(44) John LOVES, and Bill HATES, himself.

The reflexive *himself* in (44) only has a sloppy reading. However, Condition A (8) together with a multidominant analysis of Right Node Raising should predict strict readings to be available here. Why aren't they available? I have no answer.

This challenges my Condition A-based analysis of the strict readings in (35a) and (35b) because (44) is exactly the kind of example that suggests Condition A is not at work in (35a) and (35b). The alternative hypothesis would be that the reflexives in (35a) and (35b) are logophoric (Charnavel and Sportiche 2016) and thus exempt from Condition A. Then, our strict reflexives aren't necessarily referential reflexives, thus undermining the conclusion of this paper. Indeed, the reflexives in (35a) and (35b) are embedded under *picture*-NPs, environments argued to obviate a classical Condition A (Reinhart and Reuland 1993).

Nevertheless, there are examples of non-logophoric reflexives that are subject to Condition A and generate strict readings. In (45) (citation redacted), the reflexive *itself* is inanimate, making it necessarily non-logophoric: logophors refer to attitude holders, who must be animate. Still, *itself* must refer to the copy machine in both conjuncts. In (46), the reflexive *himself* is part of a PP argument of the verbs *send* and *write*. Arguments of verbs, unlike arguments of nouns, e.g., *picture*, don't host logophors (Pollard and I. A. Sag 1992; Reinhart and Reuland 1993). Yet, *himself* may refer to *John* in both conjuncts. Note that an elliptical analysis of Right Node Raising doesn't apply to (46), because English doesn't generally allow PP Ellipsis.

(45) The copy machine destroyed, though I wanted to preserve, an image of itself.

(46) John didn't send an email, nor did his wife write a letter, to himself.

Thus, the conclusions of this paper are still warranted.

4 Conclusion

I provide Right Node Raising examples which show that reflexives and pronouns in DP-sized pivots can have strict readings. DP-sized pivots can't be generated by an elliptical analysis of Right Node Raising. I show that, if we adopt a multidominant analysis to deal with these examples, we have good reasons to believe that the strict readings come from a referential reading of these reflexives and pronouns. While it is no surprise that pronouns have referential in addition to bound readings, the idea that even reflexives have referential readings runs counter to the common conception that reflexives exclusively denote bound variables. This paper provides an argument for the existence of reflexives with referential readings.

Reflexives are known as *reflexive pronouns* in more general, descriptive linguistics terms. Perhaps this terminology is more accurate than thought: reflexives are pronouns; they just also have to satisfy Condition A.

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Notes

1. Notation: Elided material is ~~struck through~~. Accented material is in UPPERCASE. Focus-marked material is denoted like _F[this]. Variable binders in the metalanguage are subscripted with the type of the variable they bind, which is rendered monospaced, e.g. λx_e . In particular, *e* is the type of entities. For any type τ , D_τ is the set of all semantic objects of type τ . Metalinguistic constants are in sans-serif, e.g. *dog*, *defended*. More notations will be introduced in the text when they first appear.
2. See Fiengo and Lasnik 1972 for an exploration of the state of affairs without this requirement in place.
3. Another Right Node Raising phenomenon that is taken to diagnose ellipsis is apparent obviation of Condition C. Consider (47): it is grammatical even though the pronoun *he* corefers with the name *John*.

- (47) He₁ hopes that Susan₂ WON'T, but the secretary knows that she₂ WILL, fire John₁ at the end of this year.

A multidominant derivation would predict (47) to be ungrammatical, because the pronoun *he* in the first conjunct does c-command the pivot. On the other hand, ellipsis is known to amnesty similar Condition C violations that would arise if the elided VP were syntactically identical to its antecedent.

Though it seems that Condition C obviation is more difficult with DP pivots (48) than with VP pivots (47), I will declare this diagnostic inconclusive.

- (48) ?? She₂ hopes [the company]₃ will fire, but they₃ are actually going to promote, Sally₂'s evil co-worker.

4. I thank a reviewer for referring me to Heim and Kratzer's condition.